

The artifact I chose for this enhancement was the Raspberry Pi thermostat that we made in our CS350 class. The original program used a couple LEDs, an LCD screen, a temperature/humidity sensor, and 3 buttons in order to simulate a very simple thermostat system. I decided to choose this project for my ePortfolio because its one of the only projects I've done where the software directly interacts with hardware and I thought it was really cool to build on it. The original project functioned as intended however it was very much designed to get a grade and wouldn't be very suitable for real world applications.

For the enhancement I used an object-oriented approach in order to redesign the application, creating the ThermostatController and ThermostatMode I am able to better manage the thermostat states. By adding these classes I was also able to improve the organization and readability of the code which would be best practice for any long term usage of the program. Additionally I added input validation so the temperature can't be set to extreme lows/highs and I was able to add exception handling for both the temperature sensor and LCD display.

This enhancement demonstrates the course outcomes by using the appropriate software engineering best practices. Through this enhancement I was able to get a better understanding of why we care about maintainability, modular design, and reliability for real world applications. The enhanced version of this code is significantly more structured and professional than the original yet it still works exactly as intended.